National University of Lesotho

Department of Physics and Electronics

EE3404: Classroom Problem Set

Semester/Year: **1/2026**
Course Title: Electrical Circuits Analysis
Instructor: Molibeli Taele (bm.taele@nul.ls)

Due date: October 21, 2026 (Wednesday),

Instructions

A. ONE question will be graded. Of course, you do not know which problems will be selected; so you should work on all of them.
B. **Late submission will not be accepted**.
C. ***Write down all the steps*** that you have done to obtain your answers. You may not get full credit even when your answer is correct without showing how you get your answer.
D. All phasor should be answered in polar form where the magnitude is positive and the phase is between -180° and 180°.
E. All sinusoid should be answered in the cosine form where the amplitude is positive and the phase is between -180° and 180°.

Questions

1) [Alexander and Sadiku, 2009, Q10.40] Find i_o in the circuit shown in Figure 1 using superposition.

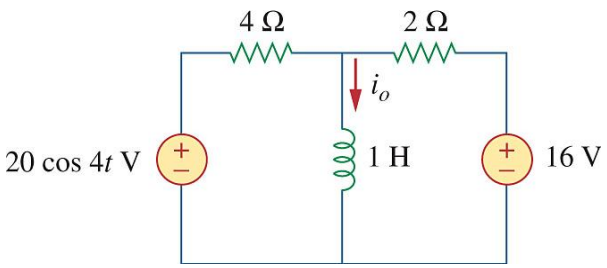


Figure 1

2) [Alexander and Sadiku, 2009, Q10.43] Using the superposition principle, find i_x in the circuit of Figure 2.

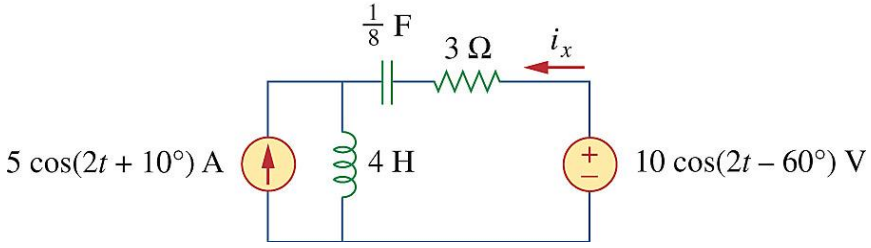


Figure 2

3) [Alexander and Sadiku, 2009, Q10.49] Using source transformation, find i in the circuit of Figure 3.

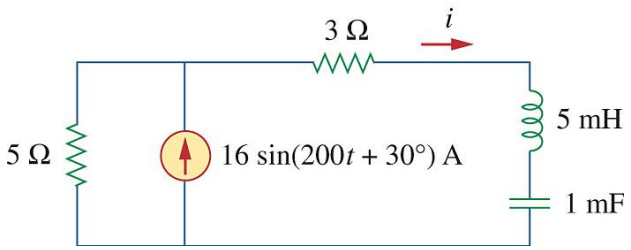


Figure 3

4) [Alexander and Sadiku, 2009, Q10.58] For the circuit depicted in Figure 4, find the Thevenin equivalent circuit at terminals a-b.

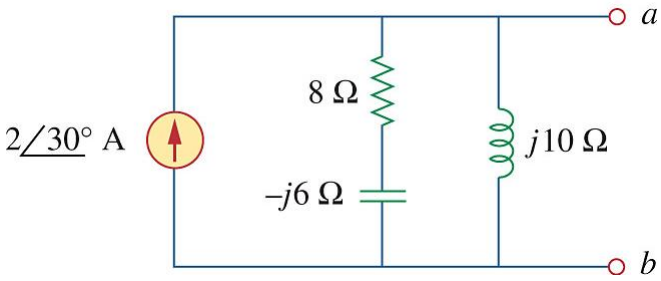


Figure 4

5) [Alexander and Sadiku, 2009, Q10.69] For the differentiator shown in Figure 5, obtain $\mathbf{V_o}/\mathbf{V_s}$. Find $v_o(t)$ when $v_s(t) = V_m \sin \omega t$ and $\omega = 1/RC$.

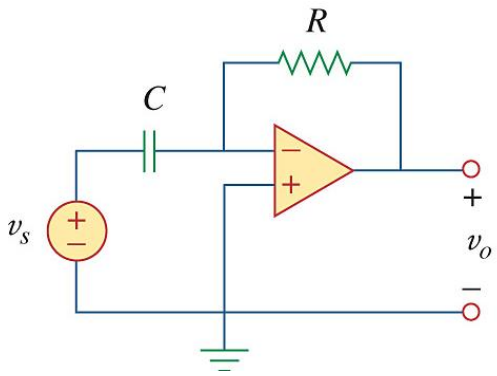


Figure 5

6) [Alexander and Sadiku, 2009, Q11.1] If $v(t) = 160 \cos 50t$ V and $i(t) = -20 \sin (50t - 30°)$ A, calculate the instantaneous power and the average power.